TrialMatch: Evidence-Grounded Patient–Trial Eligibility Screening with Cache-Aware, Tiled and Speculative Transformer Inference

Abdullah Amin
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***Abstract*—Matching a patient narrative against the inclusion and exclusion criteria of a clinical trial requires per-criterion decisions that are both correct and traceable to evidence in the record. We present TrialMatch, a 370M-parameter decoder-only Transformer (SmolLM2-360M-Instruct) fine-tuned with Low-Rank Adaptation on the TrialGPT Criterion Annotations dataset under a mandated patient-grouped 37/8/8 split. The system predicts one of six preserved expert categories per criterion—including *not enough information* and *not applicable*—and cites the supporting note sentences by identifier. Beyond task accuracy, we study inference itself: an explicitly owned key–value cache with verified greedy equivalence against an uncached reference across input/output lengths; a tiled attention kernel with online softmax validated numerically against a materialised reference and PyTorch SDPA; and a from-scratch greedy speculative decoder over a 135M/360M draft–target pair whose emitted tokens provably match plain greedy decoding. All components are exposed through a browser application with a live benchmark view. Test-set results: 63.1% accuracy and 0.342 macro-F1 after fine-tuning (base model: 32.0%), with evidence-sentence F1 of 0.212 (micro). We report honest negative results where expected—notably that a 135M draft cannot amortise its overhead against a 360M target at this scale—and analyse nine aggregate failure patterns.**

***Index Terms*—clinical decision support, patient–trial matching, LoRA, KV cache, online softmax, speculative decoding, evidence grounding**

I. Introduction and Problem Statement

Clinical trial eligibility is usually expressed as a long list of inclusion and exclusion criteria, while patient information is distributed across free-text records. Making a trial-level decision directly can hide which individual requirement is met, contradicted, or simply undocumented. TrialMatch therefore treats each criterion as a separate prediction and returns evidence sentence identifiers with the decision. Its output is a research aid for inspection, not an enrollment recommendation.

The work has two goals: measure criterion and evidence quality after lightweight supervised adaptation, and implement/test three inference mechanisms rather than only calling library switches. The contributions are (1) a fixed, patient-grouped benchmark with a constrained six-label output schema; (2) explicit cache-aware decoding, tiled online-softmax attention, and greedy speculative decoding with equivalence tests; and (3) a browser application that exposes per-criterion results, traceable sentences, and live latency comparisons. The held-out results show a useful improvement in label accuracy after tuning, but weak macro-F1 and evidence retrieval remain important limitations.

II. Related Work

TrialGPT studies patient–trial matching with large-model reasoning and retrieval over trial criteria [1]. This project instead evaluates a compact open decoder on criterion-level annotations, with a fixed split and an explicit evidence-ID output. Low-Rank Adaptation (LoRA) freezes pretrained weights and learns low-rank updates [2], reducing the trainable parameter count for a small supervised dataset. SmolLM2 is a compact open model family [3]; Qwen2.5 provides a somewhat larger candidate with a longer advertised context [4].

The Transformer uses scaled dot-product attention [5]. Exact online normalization permits softmax to be accumulated tile by tile [6]; FlashAttention demonstrates why minimizing memory traffic matters for high-performance kernels [7]. KV caching avoids recomputing keys and values for the already-seen prefix. Greedy speculative decoding uses a smaller model to propose tokens that the target verifies, preserving the target sequence while potentially reducing serial target calls [8], [9]. RoPE and grouped-query attention are relevant to the position and head layouts of modern decoder models [10], [11]. These approaches motivate an implementation and measurement study; they do not imply that an eager educational kernel will outperform fused production backends.

III. Materials and Methods

A. Dataset and Split

We use the prescribed `ncbi/TrialGPT-Criterion-Annotatior` corpus. The GPT-4 prediction columns are discarded at load time and can never enter features or metrics. Unique patient identifiers are sorted, shuffled with `numpy.random.default_rng(42)` and assigned 37/8/8 to train/validation/test (602/206/206 annotations), with

TABLE I
CANDIDATE MODEL COMPARISON USED FOR SELECTION. FP32 WEIGHT SIZES ARE PARAMETER-COUNT ESTIMATES AND EXCLUDE ACTIVATIONS AND RUNTIME OVERHEAD.

	SmolLM2-360M	Qwen2.5-0.5B
Parameters	370.5M	∼0.5B
Raw FP32 weights	∼1.48 GB	∼2.0 GB
License	Apache-2.0	Apache-2.0
Layers / Q:KV heads	32 / 15:5	24 / 14:2
Context / vocab	8,192 / 49,152	32,768 / 151,936

an executed leakage check and a published manifest of patient IDs, counts and fingerprints.

B. Model Selection

We considered SmolLM2-360M-Instruct and Qwen2.5-0.5B-Instruct as open-weight decoder-only candidates. Both have Apache-2.0 model-card licensing. The target checkpoint has 370.5M parameters (about 1.48 GB for raw FP32 weights); the alternative has 0.49B parameters (about 1.96 GB at FP32). The model cards list 8,192 and 32,768 maximum context tokens respectively, but this experiment caps combined prompt and target length at 1,024 tokens, so extra context was not the deciding factor. SmolLM2 was selected to fit the available T4/8-GB development environment with less memory pressure; it also has a 49,152-token tokenizer compatible by construction with the selected 135M draft. Evaluation uses the same target checkpoint before and after tuning. Model-card details: https://huggingface.co/HuggingFaceTB/SmolLM2-360M-Instruct and https://huggingface.co/Qwen/Qwen2.5-0.5B-Instruct. Table I summarizes the trade-off.

C. Input/Output Schema and Fine-Tuning

Each example renders the note with its *dataset-native* sentence numbering, the criterion type and text; the target is a two-line answer `LABEL: ...` / `EVIDENCE: [ids]`. Loss is computed on answer tokens only (prompt masked with -100). At inference the label slot is restricted by a character trie over the labels legal for the criterion type; unparseable output maps to *not enough information* rather than a positive decision. Prompt plus answer is truncated at 1,024 tokens; this keeps computation bounded but can remove late note content, so cited evidence quality is especially important for long records.

The final run used LoRA rank 16, alpha 32, dropout 0.05, targeting the q/k/v/o attention projections and gate/up/down MLP projections, batch size 2 with accumulation 4 (effective batch 8), learning rate 10^{-3}, three epochs, and 225 optimizer steps. There were 8.68M trainable parameters of 370.5M total. The prompt portion is excluded from loss, and training used the recorded FP32 master weights with GPU autocast. The validation loss was evaluated every 20 optimizer steps on a fixed validation subset; the training and validation curve is shown in Fig. 3. The complete run configuration is summarized in Table II.

TABLE II
FINE-TUNING CONFIGURATION OF THE FINAL LORA RUN.

Setting	Value
Train / validation examples	602 / 206
Epochs / optimizer steps	3 / 225
LoRA rank / alpha / dropout	16 / 32 / 0.05
Target modules	q/k/v/o, gate/up/down
Batch / accumulation	2 / 4
Learning rate / schedule	10^{-3} / warmup + cosine
Seq. cap / trainable params	1,024 / 8.68M
Elapsed training time	478.5 s (logged run)

D. Inference Mechanisms

1) KV cache: We own the cache as an explicit object (append, truncate, byte accounting) and drive the checkpoint's layers manually: rotary positions continue at the cached offset, prefill uses an offset-aware causal mask, decode steps are single-query attends over the full cache. The uncached reference recomputes the whole prefix per token.

2) Tiled attention: The tiled kernel walks key blocks with running maxima and normalisers (online softmax), rescaling the accumulator by $\exp(m_{\text{old}} - m_{\text{new}})$.

3) Speculative decoding: A 135M draft proposes γ tokens from its own cache; the target verifies all of them in one pass; the matching prefix is accepted, the first mismatch is corrected with the target's argmax, and a bonus token follows a full accept. Both caches are resynchronised at round boundaries; the label trie is applied to the target's argmax during verification.

E. Application Workflow

The Gradio application accepts a numbered patient note and one criterion per line (optionally prefixed `inclusion:` or `exclusion:`), loads a selected base or tuned checkpoint, and presents the predicted label, original sentence IDs, extracted sentence text, and a cautious template explanation. It derives a preliminary aggregate verdict using explicit rules: any failed criterion yields likely ineligible; otherwise unknown criteria yield undetermined; all-positive criteria yield likely eligible. This rule is not trained or scored as a ground-truth patient outcome. A downloadable JSON report records the results, timestamp, model/adapter, decoding mode and counts. A separate live benchmark view compares uncached, cached, and speculative decoding on the current note. Input validation rejects empty notes or criteria and requires numbered sentences. The interface displays an educational-use disclaimer. Start it with `cd src && python -m trialmatch.app`; the default address is http://127.0.0.1:7860.

IV. EXPERIMENTAL SETUP AND METRICS

Common protocol: deterministic greedy decoding (temperature 0), identical prompts for base vs. tuned comparisons, fixed held-out test untouched until final evaluation. Classification: accuracy, macro-F1, per-class precision/recall/F1 and confusion matrices overall and per criterion type. Evidence retrieval: micro and macro precision/recall/F1; records with an

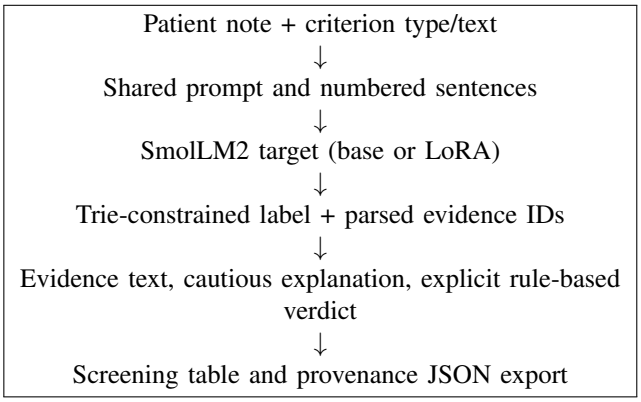


Fig. 1. TrialMatch prediction and application data flow. The derived overall verdict is a rule-based summary, not a learned or annotated label.

empty gold set are excluded from the macro mean and reported separately (an empty prediction on an empty gold set counts as agreement in the micro counts). Benchmarks: 2 warm-ups + 5 repetitions, median, `torch.cuda.synchronize()` on GPU, peak-memory capture per run.

V. Results and Discussion

A. Eligibility and Evidence Quality

The fine-tuned checkpoint reaches 63.1% accuracy and 0.342 macro-F1 on the held-out test set, compared with 32.0% and 0.097 for the base checkpoint. The tuned model emits a valid constrained label on every record, while the base evaluation has a 100% invalid-label rate; thus its fallback-based classification metrics are not a clean measure of free-form base-model label quality. On test data the tuned inclusion accuracy is 52.0% (macro-F1 0.261) and exclusion accuracy is 69.5% (macro-F1 0.273). These accuracy values mask substantial class imbalance: the test split contains 91 *not excluded* examples and zero *excluded* examples, and the train split has 387 exclusion versus 215 inclusion annotations. Consequently the zero support for the *excluded* class prevents a test-set estimate of its recall.

Evidence retrieval is much weaker than label classification: micro precision/recall/F1 are 0.281/0.170/0.212, with 16 true-positive, 41 false-positive and 78 false-negative sentence matches among records with annotated evidence. In addition, of 145 examples with no gold evidence, 12 predictions nevertheless cited sentences; those are unsupported-citation cases. Empty-gold records are excluded from macro averaging and reported separately, as specified by the evaluation convention. Fig. 2 visualizes the overall and per-type confusion matrices. It shows the exclusion model's strong preference for *not excluded* and the inclusion model's confusion between *included* and insufficient information. Table III compares overall base/tuned and split metrics; Table IV exposes the class-level behavior.

B. Correctness of the Inference Mechanisms

The cached decoder produces token-identical outputs to the uncached reference (and to HF `generate`) across 3 input/output configurations; the speculative decoder matches plain greedy on 4 configurations for $\gamma \in \{2, 4\}$. A subtle offset bug (multi-token verification queries masked by local rather than absolute positions) surfaced exactly through this suite and was fixed—evidence that token-equivalence testing is not ceremonial. The attention suite also passed eight controlled cases, including causal/non-causal masks, grouped-query heads, rectangular decode-style shapes, and a fully masked row that remains finite in the tiled kernel. The maximum observed tiled-attention absolute error in the benchmark cases was 1.07×10^{-6}; all reported cases satisfy the 2×10^{-4} test tolerance.

C. Efficiency Measurements

The cache benchmark has exact token equality in all six settings. At prompt length 1,024, cached decoding processes about 20–28 tokens/s versus 2.6 tokens/s without reuse, a roughly 7.8–11.0× decode-throughput gain; the gap grows with prefix length because full recomputation repeatedly revisits old tokens. Prefill is reported separately, and cached peak memory increases with prompt length (about 1.53–2.01 GB across the recorded cases). The settings and speedups are listed in Table V.

The tiled implementation reduces peak attention memory relative to the materialized reference at the larger sequence lengths: at $S = 2{,}048$, 32.2 MB versus 249.7 MB. However, its eager Python block loop takes 250 ms, compared with 1.89 ms for SDPA, which dispatches to a fused backend. This is the expected distinction between algorithmic memory savings and an optimized GPU kernel. Speculative decoding has a mean acceptance rate of 0.643 but achieves only 0.37–0.52× plain greedy throughput in the six measured runs. Draft model execution, verification, and resynchronization overhead outweigh the saved serial target calls at this small model scale; every output nevertheless equals target greedy exactly. The latency and memory measurements appear in Tables VI and VII.

D. Failure Analysis

Table VIII names ten representative test cases from the saved, 206-record prediction export. The most direct contradictory cases are annotations 161–164: the same patient's note describes femoral-artery catheterization, yet all four unrelated vascular-surgery inclusion criteria were labeled *included*; each prediction cited sentence 0 despite an empty gold evidence set. Annotation 165 also labels a pregnancy exclusion criterion *not excluded* rather than *not applicable*. In annotations 169 and 174, the note does not establish a pre-operative GFR or previous axillary surgery, but the model again predicts *not excluded* instead of *not enough information*. These examples show that an unmentioned exclusion condition must not be interpreted as disproven.

Evidence failures occur even when the class is correct. For annotation 179, the model predicts *not excluded* but omits gold sentence 0, which mentions cardiac catheterization. In annotation 185, it returns insufficient information with

TABLE III

CRITERION-LEVEL ELIGIBILITY CLASSIFICATION AND EVIDENCE-SENTENCE RETRIEVAL ON THE SHARED 8/8 VALIDATION/TEST PATIENT SPLITS (206 ANNOTATIONS EACH). DECODING: DETERMINISTIC GREEDY WITH TRIE-CONSTRAINED LABEL SLOT.

Model	Split	Acc. (%)	Macro-F1	Ev.-F1 (micro)	Ev.-F1 (macro)	Invalid
base	validation	25.7	0.068	0.000	0.000	1.000
base	test	32.0	0.097	0.000	0.000	1.000
tuned	validation	65.0	0.278	0.184	0.113	0.000
tuned	test	63.1	0.342	0.212	0.198	0.000

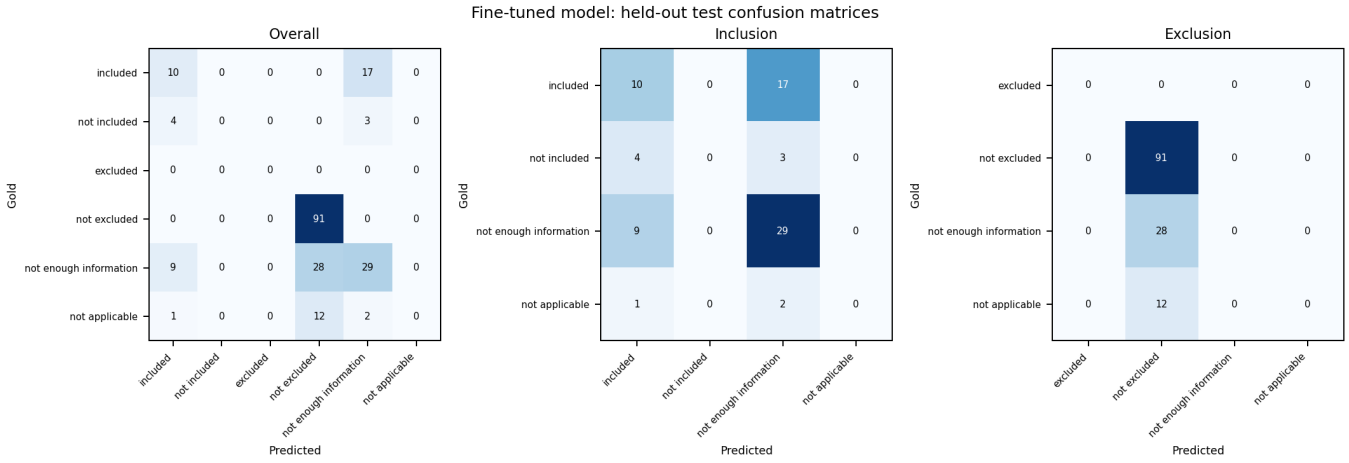


Fig. 2. Held-out test confusion matrices for the tuned checkpoint. Axes use gold rows and predicted columns; class support differs by criterion type.

TABLE IV

PER-CLASS PRECISION/RECALL/F1 OF THE FINE-TUNED MODEL ON THE HELD-OUT TEST PATIENTS (OVERALL AND BY CRITERION TYPE).

Scope	Precision	Recall	F1
overall – included	0.417	0.370	0.392
overall – not included	0.000	0.000	0.000
overall – not excluded	0.695	1.000	0.820
overall – not enough information	0.569	0.439	0.496
overall – not applicable	0.000	0.000	0.000
inclusion – included	0.417	0.370	0.392
inclusion – not included	0.000	0.000	0.000
inclusion – not enough information	0.569	0.763	0.652
inclusion – not applicable	0.000	0.000	0.000
exclusion – not excluded	0.695	1.000	0.820
exclusion – not enough information	0.000	0.000	0.000
exclusion – not applicable	0.000	0.000	0.000

TABLE V

KV-CACHE VS. UNCACHED FULL RECOMPUTE (GREEDY, IDENTICAL INPUTS; DEVICE CUDA, MEDIAN OF 5 AFTER 2 WARM-UPS).

S	new	cached tok/s	uncached tok/s	speedup	equal
128	16	28.63	19.47	1.5×	True
128	48	28.07	16.53	1.7×	True
512	16	28.58	5.87	4.9×	True
512	48	27.68	5.50	5.0×	True
1024	16	19.94	2.55	7.8×	True
1024	48	28.02	2.56	11.0×	True

TABLE VI

FORWARD-PASS LATENCY OF ONE ATTENTION LAYER (CUDA, FP32).

S_q	heads	reference (ms)	tiled (ms)	SDPA (ms)
256	6	0.44	7.53	0.14
512	6	0.56	22.84	0.29
1024	8	1.92	64.48	0.73
1024	8	3.58	64.73	1.25
2048	6	5.73	249.97	1.89

no citations, although the annotation marks sentences 2–3 (tachypnea/tachycardia and hypotension) as evidence for the shock/ARDS inclusion criterion. Annotation 193 likewise predicts *not excluded* for "Patients were moribund" and cites nothing despite the gold annotation spanning all five note sentences. Latency outliers 307, 831, and 832 take 1.177–1.231 s; examples 307 and 831 over-cite sentences 0–4 while omitting the gold sentence 3 or 7, and 832 cites sentences 0–4 despite an empty gold set. Across the full test export, overlapping categories include 39 wrong-label/evidence- correct, 37 wrong-label/evidence-wrong, 40 missing-evidence, 12 unsupported-evidence, 37 insufficient-information, four contradictory-verdict and three latency-outlier records. The trie constrained every label to its criterion type; the main remaining risks are unsupported affirmations and evidence localization, not invalid label syntax. These categories overlap: for example, annotations 161–164 count as contradictory verdicts, unsupported citations, and wrong-label/evidence-wrong cases. The exported error file contains representative examples; category totals are computed from the complete prediction set.

TABLE VII
GREEDY SPECULATIVE DECODING ($\gamma = 4$) VS. PLAIN GREEDY ON THE SAME PROMPTS; DRAFT SMOLLM2-135M, TARGET SMOLLM2-360M.

prompt	greedy tok/s	spec tok/s	speedup	acc. rate	rounds	equal
180	19.41	10.14	0.52×	0.64	7	True
180	26.20	12.62	0.48×	0.64	7	True
180	26.52	12.85	0.48×	0.64	7	True
180	26.61	13.22	0.50×	0.64	7	True
180	26.58	9.81	0.37×	0.64	7	True
180	26.43	13.20	0.50×	0.64	7	True

TABLE VIII
SELECTED CASE-LEVEL ERRORS AND OUTLIERS FROM THE TUNED HELD-OUT TEST PREDICTIONS. EVIDENCE IDS REFER TO THE ORIGINAL NUMBERED PATIENT NOTE.

ID	Type	Criterion	Gold → prediction	Evidence IDs (gold; prediction)	Latency (s)
161	inclusion	Patients undergoing elective carotid endarterectomy	not included → included	gold []; pred [0]	0.345
165	exclusion	Pregnancy	not applicable → not excluded	gold []; pred []	0.725
169	exclusion	Patients with an estimated pre-operative glomerular filtration rate ¡...	not enough information → not excluded	gold []; pred []	0.405
176	inclusion	No medical history of central vena catheterization.	included → not enough information	gold [0]; pred []	0.351
179	exclusion	Had been performed central venous puncture or catheterization before.	not excluded → not excluded	gold [0]; pred []	0.302
185	inclusion	Patients were included if they were diagnosed with Shock, Acute respiratory...	included → not enough information	gold [2, 3]; pred []	0.363
193	exclusion	Patients were moribund.	not enough information → not excluded	gold [0, 1, 2, 3, 4]; pred []	0.295
307	inclusion	Patient selection should be cautious, covering basically 5 types of liver...	not enough information → not enough information	gold [3]; pred [0, 1, 2, 3, 4]	1.177
831	inclusion	Oxygen saturation of ¿ 94 % and normal blood pressure (Systolic between 150 -...	not enough information → not enough information	gold [7]; pred [0, 1, 2, 3, 4]	1.231
832	inclusion	Symptom Score no greater than 6 (out of a possible 24) for total symptom score...	not enough information → not enough information	gold []; pred [0, 1, 2, 3, 4]	1.200

VI. LIMITATIONS

The 360M model is small for complex clinical reasoning, the study uses one dataset and one patient-grouped split, and the held-out test has zero examples of the *excluded* label. Evidence recall is low, and template explanations summarize predicted citations rather than independently validating clinical reasoning. The patient-level verdict is rule-based and has no ground-truth label. Evaluation uses deterministic greedy decoding; benchmarks are hardware- and implementation-specific. Long notes may be truncated at 1,024 tokens. The tuned adapter and per-example test/validation exports are now present locally. The adapter is excluded from Git but is available with the prediction exports and compiled report from the public artifact release. Application screenshots are included in the repository; the separately recorded 5–7-minute demonstration link remains outstanding.

VII. CONCLUSION

TrialMatch demonstrates criterion-level eligibility screening with constrained outputs, sentence-ID evidence, LoRA adaptation, and from-scratch cache-aware, tiled-attention, and speculative-decoding paths. LoRA improves held-out accuracy over the same base model, while macro-F1 and evidence retrieval reveal uneven class behavior and weak localization. The cache and speculative implementations pass greedy-token equivalence; the tiled kernel matches the reference numerically while trading speed for reduced memory. Measurements also show that a correct speculative decoder can be slower when draft overhead dominates. The main defense takeaway is to distinguish correctness of the inference algorithms from task quality and from real-world clinical validity.

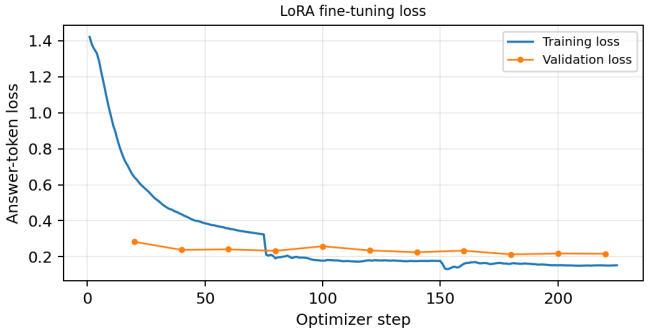


Fig. 3. Training and fixed-subset validation answer-token loss from the final recorded LoRA run.

AI-USE APPENDIX

AI assistance was used for code drafting and debugging, report organization, and editing. The submitted author is responsible for verifying and understanding the implementation, results, sources, and any course-specific disclosure requirements. The model outputs are experimental results, not clinical advice.

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